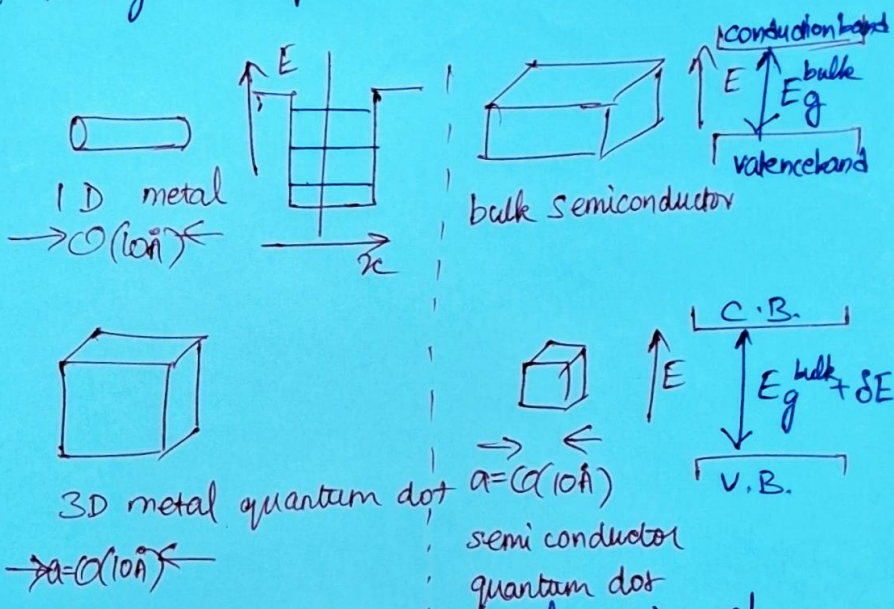


28th March' 2026

## Quantum dots

- In particle in a 1D box, energy levels are quantized due to boundary conditions
- In 3D box, the boundary conditions lead to energy level quantization. This is called quantum dot.
- In the case of semiconductors, the energy band gap is a material property. However, the energy band gap of a semiconductor quantum dot changes compared to bulk counterpart.

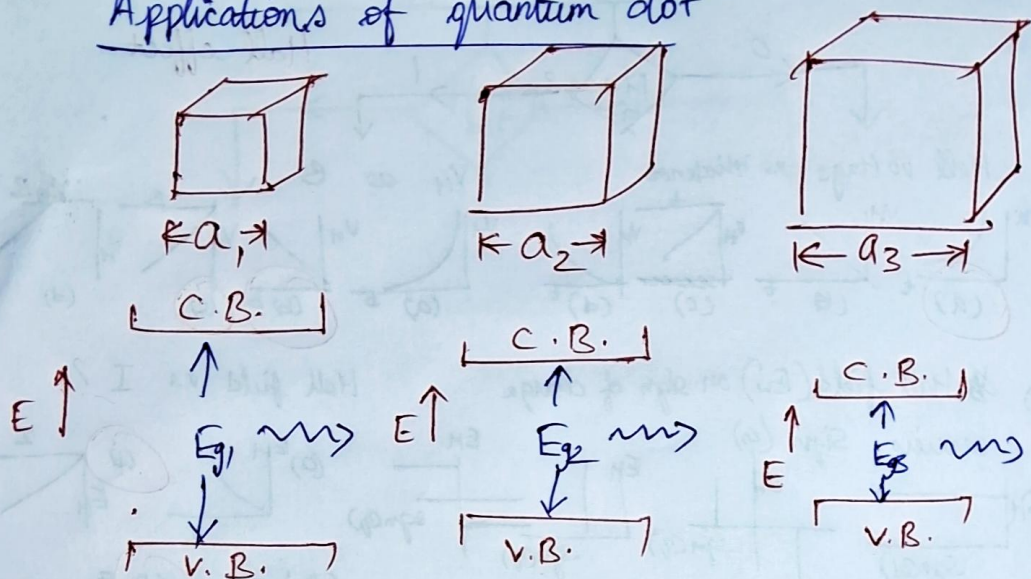


The band gap increases as the size of the quantum dot decreases

$$E_g^{\text{quantum dot}} = E_g^{\text{bulk}} + \Delta E$$

$$\Delta E \propto \frac{1}{a}$$

# Applications of quantum dot



Here  $a_1 < a_2 < a_3$

So  $E_{g1} > E_{g2} > E_{g3}$

Since light emitted by a semi conductor has frequency given by energy band gap as

$$E = h\nu,$$

the frequency of photon emitted by Q.Dots has the relation

$$\nu_1 > \nu_2 > \nu_3$$

Since wavelength is related to frequency by

$$\lambda = \frac{c}{\nu},$$

the wavelength of photon emitted by Q.Dots has the relation

$$\lambda_1 < \lambda_2 < \lambda_3$$

If  $\lambda_1 = \text{blue}$ ,  $\lambda_2 = \text{green}$ ,  $\lambda_3 = \text{red}$ , we can create Q.Dot based pixel. This is the principle in Q.Dot LED displays.